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A Comparative Study Of Super-Resolution Interpolation Techniques: Insights For Selecting The Most Appropriate Method

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Abstract

Super-resolution interpolation is a popular technique, which is used to increase the image's resolution beyond its original size. However, there are several interpolation techniques available for super-resolution, and determining which technique to use for a given image can be challenging. The aim of the project is to perform a comparative study of different interpolation techniques for super-resolution and identify the best technique for different images. It starts by collecting a dataset of images with different characteristics such as noise, blur, and contrast and then preprocess the images and apply different interpolation techniques such as nearest-neighbor, bilinear, bicubic, Lanczos and Spline etc. The super-resolved images are evaluated and compared using different metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Mean Opinion Score (MOS). Based on the results of comparative study, conclusions about the strengths and weaknesses of each method are drawn. And the most appropriate interpolation technique for specific application is identified .

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Keywords: BiLinear; Nearest Neighbour; BiCubic; Lanczos; Spline; Super-Resolution

1. Introduction

Super-resolution interpolation aims to boost an image's resolution beyond its original dimensions, thereby improving its quality and making it suitable for various applications. Numerous interpolation techniques are available for super-resolution, each with its strengths and weaknesses. Choosing the most appropriate technique for a given image can be challenging, as it depends on various factors, such as noise, blur, and contrast. Therefore, it is essential

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to perform a comparative study of different interpolation techniques to determine the most suitable method for specific applications. The paper presents a comparative study of different interpolation techniques for super-resolution interpolation. A dataset of images with different characteristics are collected, preprocessed and are applied various interpolation techniques, such as nearest-neighbor, bilinear, bicubic, Lanczos, and Spline, to obtain super-resolved images. The quality of the super-resolved images is evaluated using different metrics, such as PSNR, SSIM, and MOS. Based on the comparative study, conclusions about the pros and cons of each method are drawn and identified the most appropriate interpolation technique for specific application. The results can be useful in guiding researchers and practitioners to choose the most appropriate interpolation technique for their specific needs.

1.1. Motivation

The motivation for the project stems from the fact that super-resolution interpolation has innumerable practical applications in fields such as medical imaging, remote sensing, and surveillance. However, determining the best interpolation technique for a given image can be challenging due to the availability of various methods. Therefore, there is a need for a comparative study of different interpolation techniques to identify the most appropriate method for specific applications. The paper aims to perform such a study by collecting a dataset of images with different characteristics and evaluating the quality of the super-resolved images obtained using different interpolation techniques. The results of the study will provide valuable insights into the weaknesses and strengths of each method, and help in identifying the best interpolation technique for specific applications.

1.2. Scope

The study is limited to a specific set of image characteristics, such as noise, blur, and contrast, and does not encompass all possible image variations .The scope of the project will be limited to the evaluation of super-resolution interpolation techniques and will not include the development of new techniques

1.3. Objectives

- To preprocess the images by removing any artifacts or noise and resizing them.
- To implement and compare different interpolation techniques, including nearest-neighbor, bilinear, bicubic, Lanczos, and Spline.
- To evaluate the quality of the super-resolved images using different metrics, including PSNR, SSIM, and MOS, and compare the results.
- To offer suggestions for choosing the best interpolation method for various images, on the basis of the results of the evaluation

1.4. Organization

The paper is organized as follows: The literature study of numerous works on different interpolation techniques is covered in the first section of Section 2.The next section, Section 3, which comes next, goes into further detail on the suggested strategy and design. In Section 4, comparison is made among the several techniques .The results are discussed in Section 5.

2. Related Works

[1] presents a novel approach to image interpolation that incorporates covariance-based adaptation. The algorithm is composed of several key steps. Firstly, the local covariance coefficients are estimated from a low-resolution image. The estimation process provides insights into the statistical relationships between neighboring pixels. Subsequently, using these covariance estimations, the interpolation is adjusted to a greater resolution. The geometry of the low-resolution and high-resolution covariances is the foundation for the adaptation..By exploiting this duality, the algorithm is able to enhance the interpolation results, particularly in areas with prominent edges. To strike a balance

between computational complexity and performance, the paper proposes a hybrid approach that dynamically switches between two interpolation methods: covariance-based adaptive interpolation and bilinear interpolation . Bilinear interpolation is a computationally efficient method commonly used in image processing. However, in areas with significant edge information, bilinear interpolation may result in suboptimal results. By integrating the covariance-based adaptive interpolation, the algorithm can overcome the mentioned limitation and improve the quality of the interpolated images. **Advantages:** By adaptively adjusting interpolation coefficients based on local covariance estimates, finer details are captured, resulting in visually improved images. The mathematical duality between low-resolution and high-resolution covariances is taken advantage of by the procedure. **Disadvantages:** In this process, colour picture reconstruction from CCD data and resolution augmentation of grayscale images are explored. It also explores algorithm's effectiveness in other image processing tasks, such as denoising or super-resolution, remains unexplored and requires further investigation.

The limitations of traditional image enlargement algorithms, such as nearest neighbor interpolation, bilinear interpolation, and high-order interpolation are addressed in [2] and introduced a hybrid algorithm that combines deep learning and traditional interpolation techniques. The proposed algorithm consists of two primary steps. In the first step, deep learning is employed to perform the reconstruction of low-resolution data. Deep learning algorithms excel at automatically extracting features from data, allowing for enhanced reconstruction. By leveraging the power of deep learning, the algorithm can generate improved low-resolution reconstructions. In the second step, the results obtained from the traditional interpolation reconstruction are combined with the low-resolution reconstructions achieved through deep learning. Deep learning algorithms are then utilized again to further train and refine the combined results, ultimately yielding high-resolution reconstructed data. The two-stage approach allows for the fusion of the strengths of both traditional interpolation techniques and deep learning algorithms. Moreover, the proposed algorithm excels in reconstructing fine details, resulting in clear outlines and high-quality images. By leveraging the complementary capabilities of traditional interpolation and deep learning, the algorithm achieves superior super-resolution reconstruction. **Advantages:** By integrating traditional interpolation methods with deep learning, the algorithm leverages the strengths of both approaches. . **Disadvantages:**The dataset used in the study was limited to specific medical fields, which may limit the generalizability of the proposed method to other applications and domains.

The loss of image's quality is frequently the outcome of image resampling, such as when rotating an image. To mitigate this issue, the choice of an appropriate interpolating function is crucial. To evaluate the interpolating functions, [3] compares their frequency responses. Functions that extend over four picture elements (Eg: cubic B-spline) exhibit better frequency responses compared to functions that extend over one (nearest neighbor) or two (linear) pixels. The paper also analyzes the effects of each interpolating function on the image quality. While linear and cubic B-spline interpolation attempt to smooth the image, the nearest neighbour function shifts it up to half a pixel. The high-resolution cubic spline functions outperforms, indicating their superior ability to preserve image details during resampling. Also [3] highlights the importance of resampled points' location, relative to original coordinate system. When the points are aligned, the data are precisely reproduced, while equidistant resampled points result in the most smoothing of the image. **Advantages:** The paper systematically compares and evaluates five interpolating functions, providing insights into their frequency responses and performance during image resampling. The analysis demonstrates that cubic spline functions (high-resolution) exhibit the best performance and preserve image details effectively, making them a suitable choice for image resampling tasks. **Disadvantages:** The paper specifically examines the five mentioned interpolating functions, potentially limiting the generalizability of the findings to other interpolation methods. The high-resolution cubic spline functions, while providing superior image quality, require additional computing time which may impact real-time applications.

[4] presents a comparative analysis of various interpolation techniques commonly used in processing a image. It highlights that bicubic interpolation outperforms the methods of interpolation, nearest neighbor and bilinear , in terms of conserving details of image. Bicubic interpolation provides smoother results and captures finer details compared to the other two techniques. This makes it a preferred choice for applications where preserving subtle information is crucial. Furthermore, the paper investigates the effectiveness of more advanced techniques, namely DWT and Kriging, in preserving image details. The results indicate that both DWT and Kriging interpolation techniques offer finer details compared to the traditional methods. These techniques leverage the power of wavelet analysis and geo statistics, respectively, to improve the accuracy of interpolated images. **Advantages:** The article provides a detailed comparison of different interpolation techniques with reference to accuracy, computational complexity as well as visual quality.

Disadvantages: The paper provides an overview of various interpolation techniques, but it does not provide specific guidance on technique selection for different application scenarios

The generalization of weighted FRI techniques by enhancing two alternative representative FRI methods is focused in [5].The generalized weighted FRI algorithms presented in the paper extend the conventional unweighted FRI methods by incorporating individual attribute weights. By integrating attribute weights throughout the FRI procedures, the algorithms can derive interpolated conclusions that account for the varying significance of different antecedent attributes. This enhancement allows for more accurate and nuanced interpretations of observations that do not precisely match any given rules. The paper reports the results of systematic comparative evaluations performed on benchmark classification problems. The empirical evaluations demonstrate the effectiveness and efficiency of the generalized weighted FRI algorithms using only two nearest neighboring rules. The algorithms effectively handle interpolation tasks and exhibit improved performance compared to conventional unweighted FRI methods. These findings highlight the potential of attribute-weighted FRI techniques in enhancing the interpretation and classification capabilities of fuzzy rule-based systems. **Advantages:** It improves interpretability of fuzzy rule-based systems by incorporating attribute weights. Considering attribute weights, leads to improved classification performance, particularly in scenarios where conventional unweighted methods may struggle to provide meaningful results. **Disadvantages:** The paper's focus on generalizing weighted FRI techniques is limited to two representative FRI methods and their corresponding procedures.

3. Methods

3.1. Architecture

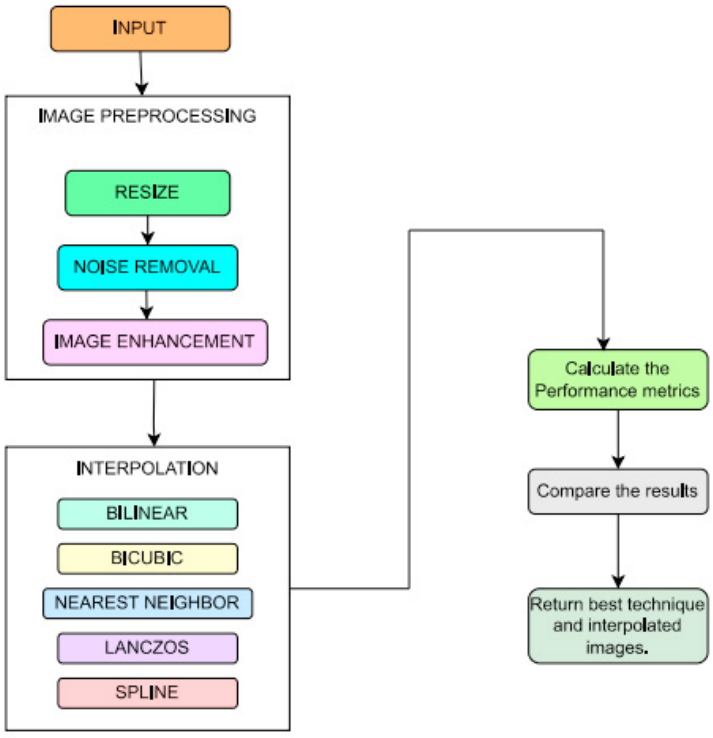
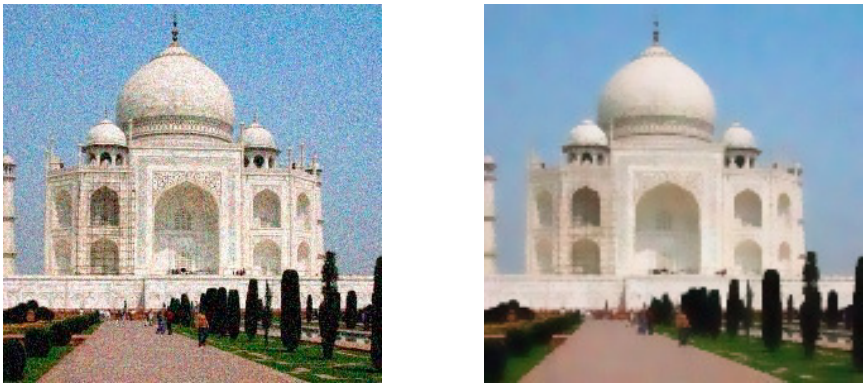


Fig. 1. Proposed Model

Figure 1 depicts the proposed model for selecting the most appropriate interpolation method for a given image.

3.2. Preprocessing

- Image Resizing: Resize the input image
- Noise Removal : Images can be influenced by noises such as salt & pepper noise,gaussian noise, and speckle noise[6]. Techniques like Median Filtering and Gaussian Filtering can remove salt and pepper noise, but Bilateral Filtering can successfully reduce Gaussian and salt and pepper noise[7].
- Image Enhancement: Noise removal techniques can be used to enhance image quality. Image enhancing techniques such as False Colour Composite, on the other hand, can be used to recover lost data. It assigns different colors to different electromagnetic spectrum bands.



(a) Image before Preprocessing (b) Image after Preprocessing

Fig. 2. Preprocessing

3.3. Nearest Neighbor Interpolation

Nearest Neighbor interpolation is a simple yet widely used technique in image processing and computer graphics for resizing or rescaling images. In this technique, the value of a new pixel is decided by the value of the closest neighboring pixel in the original image. It is straightforward to implement and computationally efficient, making it suitable for real-time applications and scenarios where speed is crucial. When scaling up an image, each pixel in the original image is replicated to fill multiple pixels in the resized image. Each new pixel is assigned a value equal to the closest pixel's value in the original image. The same can be visualized as a grid where each new pixel is placed at the center of a corresponding region in the original image. Mathematically, the formula for Nearest Neighbor interpolation can be formulated as:

$$\text{NewPixel}(x_{\text{new}}, y_{\text{new}}) = \text{OriginalPixel}(\text{round}(x_{\text{new}}), \text{round}(y_{\text{new}}))\text{.....}(1)$$

. Here, $(x_{\text{new}}, y_{\text{new}})$ represents new pixel's coordinates in the resized image, and round() denotes rounding the coordinates to the nearest whole number. By rounding the coordinates, the nearest neighbor in the original image is determined. From the study of [8] and [9] , we understood that, Nearest Neighbor interpolation works well when preserving sharp edges and pixel-level details is essential. Since it directly copies the value of the nearest pixel, it can maintain sharp boundaries without introducing blurring or smoothing effects. Consequently, it is suitable for simple graphics, pixel art, or cases where computational efficiency is crucial. However, Nearest Neighbor interpolation has limitations. It tends to produce jagged or blocky artifacts when scaling up images significantly or when rescaling images with fine details. The lack of smoothing or interpolation between pixels can result in a loss of image quality and visual fidelity, particularly in photographs or continuous-tone images. In such cases, more advanced interpolation techniques like bilinear or bicubic interpolation are preferred to achieve better image quality and smoother transitions. Let's imagine that the requirement is to upscale a 3x3 image by a factor of 2, as demonstrated below. Take a look at

pixel in the unknown image (right side matrix). To give it a value, the input 3x3 image's nearest pixel (left side matrix) is to be located.

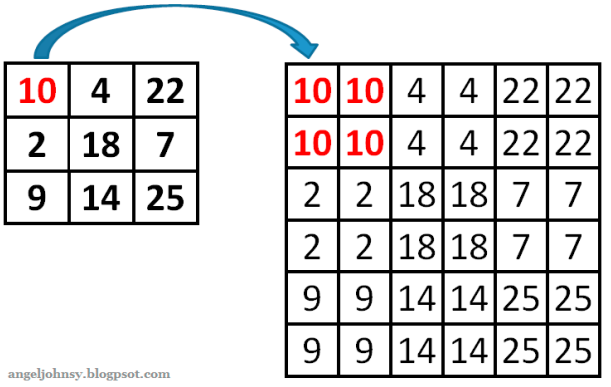


Fig. 3. Nearest Neighbor interpolation

Figure 3: Nearest Neighbour Interpolation [9] Nearest neighbor interpolation is a good choice for applications where speed is critical, such as real-time video processing.

3.4. BiLinear Interpolation

The term "bi-linear interpolation" refers to using linear interpolation in two directions. The result is produced using the weighted average of the four closest neighbours. From the experimental results (figure 9a and 10a) obtained and the study of [1] and [3], Bilinear interpolation is particularly well-suited for images that have gradual color or intensity changes. It works effectively for natural images, photographs, and continuous-tone images. It preserves the overall structure and smoothness of the image while reducing artifacts caused by aliasing. Bilinear interpolation is commonly used in applications such as image resizing, zooming, and rotation. However, it is important to note that bilinear interpolation may not perform as well when there are sharp edges or fine details in the image. It can introduce blurring effects or lose some of the high-frequency information present in the original image. In such cases, more advanced interpolation techniques, such as bicubic interpolation or Lanczos interpolation, may be preferred.The following equation covers how weights are distributed in OpenCV.

$$fx = (float)((dx + 0.5) * scale_x - 0.5).....(2)$$
$$sx = cvFloor(fx)$$
$$fx- = sx$$

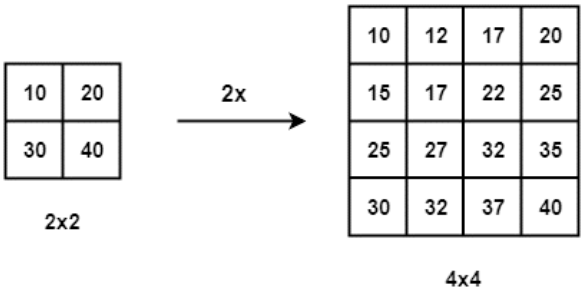


Fig. 4. BiLinear interpolation

The left pixel is given a weight of 1-fx where dx is the column index of the unknown pixel and fx is the weight provided to the right pixel. The output width divided by the input width is known as $scale_x$. The row index for y is dy, and the current ratio of heights is $scale_y$. After knowing how weights are calculated, the following are the results. Bilinear interpolation is a good choice for applications where accuracy is important, but speed is not critical, such as image resizing.

3.5. Bicubic Interpolation

Bilinear interpolation assumes a linear transition of pixel values between neighboring pixels because of which, it may produce artifacts and blurring effects when applied to images with sharp edges, fine details, or high-frequency components. Instead of using a square neighborhood, bicubic interpolation uses a larger rectangular neighborhood of 16 pixels (4x4) around the target pixel.In bicubic interpolation, the new pixel value is determined by evaluating a cubic function based on the 16 neighboring pixels. From the study of [10] and [11] and visual experimental result (figure 9 b) , the advantage of bicubic interpolation is its ability to handle images with sharper edges and fine details. It can better preserve the sharpness and clarity of edges compared to bilinear interpolation. Bicubic interpolation also tends to produce more visually pleasing results when upscaling images or resizing them to a larger size. However, it's important to note that bicubic interpolation may introduce some blurring effects in certain situations. If the image contains very fine details or high-frequency components, bicubic interpolation may not be able to capture all the intricate details accurately. In such cases, more advanced interpolation techniques, such as Lanczos or Spline interpolation, may be more suitable. The following code specifies how weights are distributed in OpenCV.

const float A = -0.75f;

coeffs[0] = $((A \cdot (x+1) - 5A) \cdot (x+1) + 8A) \cdot (x+1) - 4A$;

coeffs[1] = $((A+2)x - (A+3))x^2 + 1$;

coeffs[2] = $((A+2)(1-x) - (A+3))(1-x)^2 + 1$;

coeffs[3] = 1 − coeffs[0] − coeffs[1] − coeffs[2];

x used in the above code is calculated from below code where x = fx

$$fx = (float)((dx + 0.5) * scale_x - 0.5) \quad \text{.....(3)}$$

$$sx = cvFloor(fx)$$

$$fx- = sx$$

Similarly, for y, replace x with fy and fy can be obtained by replacing dx and $scale_x$ in the above code by dy and $scale_y$ respectively

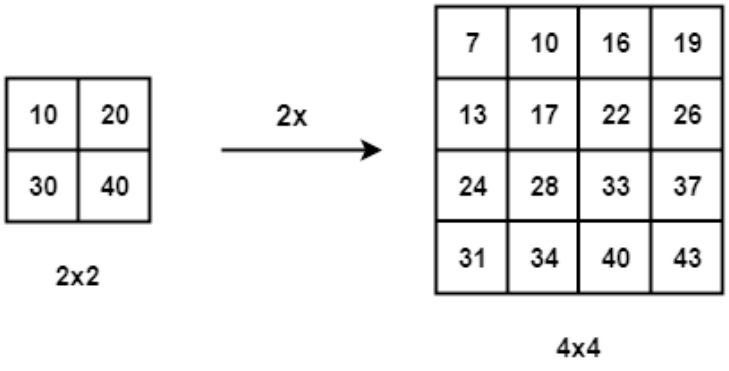


Fig. 5. Bicubic interpolation

Bicubic interpolation is a good choice for applications where both accuracy and speed are important, such as image smoothing.

3.6. Lanczos Interpolation

While bicubic interpolation provides good results in many cases, it can introduce certain artifacts such as blurring or ringing effects. Lanczos interpolation, on the other hand, aims to address some of the limitations of bicubic interpolation. It is based on a mathematical windowing function known as the Lanczos kernel, which is used to calculate the weighted average of pixels during the resizing process. The Lanczos kernel has the property of being a ”sinc” function, which means it has oscillations and decays towards zero. These oscillations help to preserve sharp edges and fine details in the image during resizing. By using a larger number of oscillations in the kernel, Lanczos interpolation can provide better preservation of image details compared to bicubic interpolation. From the comparative study provided in [12] and [13] it is clear that ,Lanczos interpolation is particularly well-suited for images with sharp edges, high-frequency details, and fine structures, such as text or line art. It can help to minimize the blurring effect often seen in bicubic interpolation, resulting in sharper and more accurate representations of these types of images.However, it’s important to note that Lanczos interpolation may not be suitable for all types of images. In some cases, especially for images with smooth gradients or continuous color transitions, the oscillations of the Lanczos kernel can introduce unwanted artifacts such as ringing or halo effects. In these situations, other interpolation methods such as bicubic interpolation or bilinear interpolation may be more appropriate.

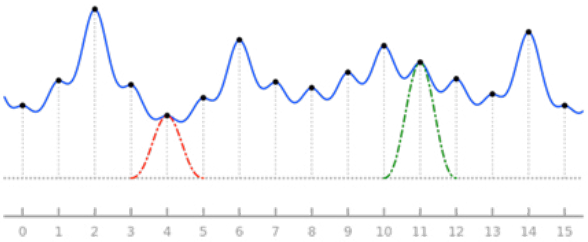


Fig. 6. Lanczos Interpolation

Figure 7 shows the partial plot of a discrete signal with size parameter an equal to 1 and resulting Lanczos interpolation (that is, solid blue curve). Two replicas of the Lanczos kernel that have been scaled and shifted to correspond to samples 4 and 11 are also displayed (dashed curves). Lanczos interpolation is a good choice for applications where accuracy is critical, and speed is not critical, such as image denoising.

3.7. Spline Interpolation

Lanczos interpolation uses a windowed sinc function to estimate the missing pixel values. However, it can introduce ringing artifacts, especially in high-contrast images, due to the nature of the sinc function. In contrast, spline interpolation offers a smoother and more visually pleasing result, making it suitable for a wide range of image types. Spline interpolation works by fitting piecewise polynomials, called spline curves, to the data points in the image. These curves are designed to be smooth and continuous, ensuring that the interpolated values blend well with the existing data. The main advantage of spline interpolation lies in its ability to handle both low-frequency and high-frequency details in an image. Low-frequency information, such as smooth gradients and color transitions, is accurately preserved by the spline curves. At the same time, high-frequency details, such as sharp edges and fine textures, are also preserved due to the smoothness of the interpolated result. From [14] and [15] , Spline interpolation is particularly effective when used with images that have complex structures, intricate patterns, or a combination of smooth and sharp features. It excels at preserving fine details and avoiding undesirable artifacts like jagged edges or halos. Moreover, spline interpolation is used in various applications like resizing an image, zooming, and image editing, where maintaining image quality and fidelity are crucial. However, it’s worth noting that spline interpolation might not be the most suitable choice for all scenarios. In cases where preserving sharp edges or pixel-level accuracy is of utmost importance, other interpolation methods, such as nearest neighbor or bilinear interpolation, may be preferred. Additionally, spline interpolation can be computationally more intensive compared to simpler interpolation techniques.

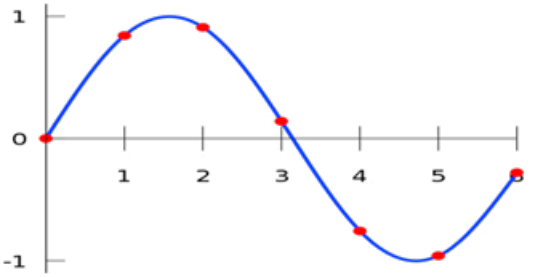


Fig. 7. Spline Interpolation

Figure 8 depicts the Plot of data with spline interpolation applied. Spline interpolation is a good choice for applications where accuracy is important, and speed is not critical, such as 3D surface reconstruction.

3.8. Algorithm

Algorithm 1

Input: Image or path of image X

Output: Best suitable technique and images interploated using 5 techniques X'

1: Input the image or path of image from the user.X

2: Prepare of list of interpolation techniques. X'

3: Resizing: As the first step of preprocessing image, apply resize function in openCV to resize the image .

4: Noise Removal: Apply bilateral filter function to remove the noise in the image.

5: Image Enhancement: False Color Composition technique was used for image enhancement.

6: Apply BiLinear Interpolation

$$f(x, y) = (1 - dj)(1 - di)f(i, j) + (1 - dj)dif(i + 1, j) + dj(1 - di)f(i, j + 1) + djdif(i + 1, j + 1)...(4)$$

Where f(x,y) is a function defined on a rectangular grid of values with rows labeled by i and columns labeled by j. Compute the value of f at a point (x,y) that lies within the rectangle spanned by (i,j), (i+1,j), (i,j+1), and (i+1,j+1). Here di and dj are the distances between x and i, and y and j, respectively.

7: Apply BiCubic interpolation

$$p(x, y) = \sum_{i=0}^{3} \sum_{j=0}^{3} a_{ij} x^i y^j ...(5)$$

Where p(x, y) represents the interpolated value at a specificlocation (x, y) in the dataset. Bicubic interpolation aims to estimate this value based on the surrounding grid points. aij are the coefficients or weights associated with each grid point. There are 16 coefficients in total, corresponding to the 4x4 grid of nearby points. x, y represents the fractional distance between the target point and the nearest grid point in the x-direction, y-direction respectively.

8: Apply Nearest Neighbor interpolation

I'(X', Y') = I(round(X'), round(Y'))...(6).

Where (x',y') be the coordinates of a pixel in the new image that are to be interpolated and I(x,y) be the intensity value of the pixel at coordinates (x,y) in the original image

9: Apply Lanczos interpolation

Given a one-dimensional signal with samples si, for integer values of i, the value S(x) interpolated at an arbitrary real argument x is obtained by the discrete convolution of those samples with the Lanczos kernel:

$$S(x) = \sum_{i=|x|-a+1}^{|x|+a} S_i L(x - i)...(7)$$

where a is the filter size parameter, and x is the floor function. The bounds of this sum are such that the kernel is zero outside of them.

Algorithm 2 Algorithm 1 (cont.)

10: Apply Spline interpolation The spline interpolation function $S(z)$ is defined as follows:

$S(z) = S_i(z), z_i \leq z \leq z_{i+1}$...(8)

where $S_i(z)$ is the cubic polynomial defined on the interval $[z_i, z_{i+1}]$. The cubic polynomial $S_i(z)$ can be written in the form:

$S_i(z) = a_i + b_i(z - z_i) + c_i(z - z_i)^2 + d_i(z - z_i)^3$...9

where a_i, b_i, c_i, and d_i are constants that depend on the data points.

11: Compare the resultant images of each technique by means of calculating quality metrics (PSNR, SSIM, MOS)

12: Use the calculated metrics and suggest best technique for the given image

4. Comparision of outputs obtained

The section presents a comprehensive analysis of a comparative study conducted on various super-resolution interpolation techniques. The primary objective of the study was to gain insights into selecting the most appropriate method for enhancing the resolution of low-resolution images. Through rigorous experimentation and analysis, the section aims to present the findings, discuss the results, and provide valuable conclusions that will aid researchers and practitioners in making informed decisions regarding the choice of super-resolution interpolation techniques.



Fig. 8. Original image and image after Bilinear interpolation

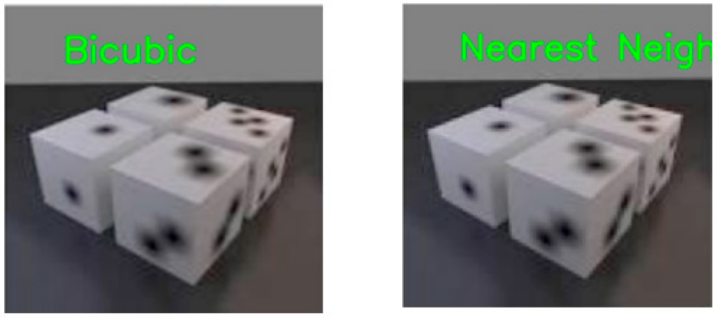


Fig. 9. Images after Bicubic and Nearest neighbor interpolation

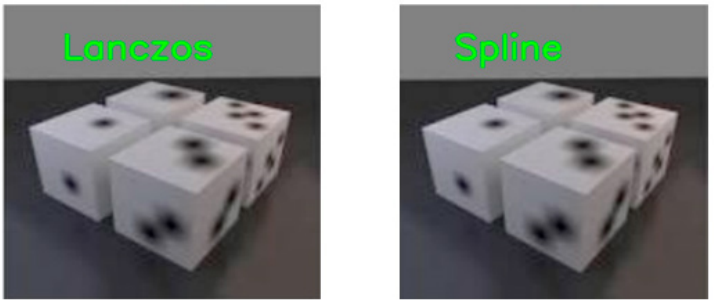


Fig. 10. Images after Lanczos and Spline interpolation

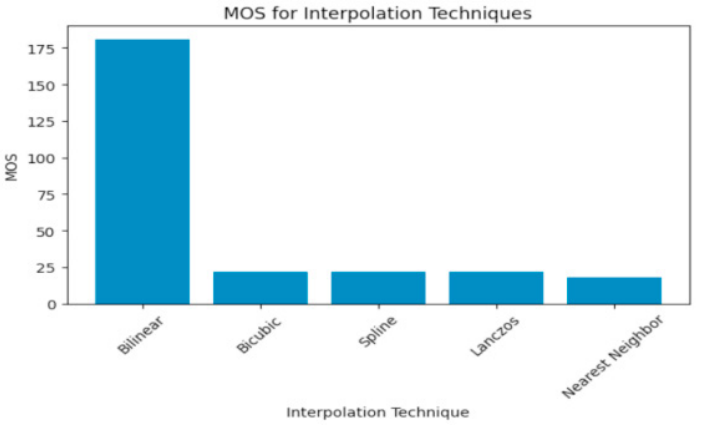


Fig. 11. Bar graph of MOS metrics for the interpolated images of original image in Figure 8



Fig. 12. Original image and image after Bilinear interpolation



Fig. 13. Images after Bicubic and Nearest Neighbour interpolation



Fig. 14. Images after Lanczos and Spline interpolation

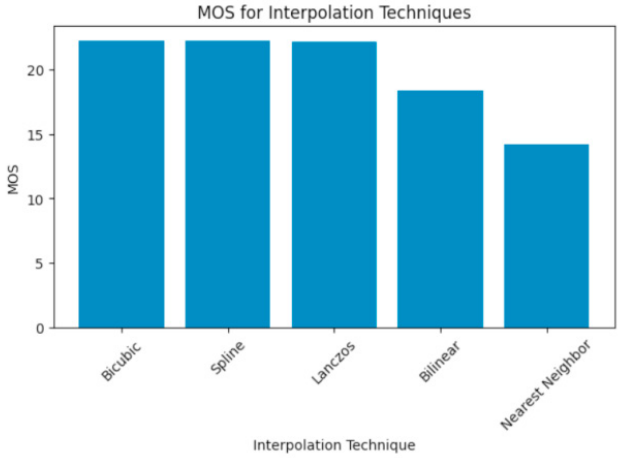


Fig. 15. Bar graph of MOS metrics for the interpolated images of original image in Figure 12

5. Results and Discussion

The following are the results obtained after interpolating the images 8(a) and 12(a) with various techniques . Based on the performance metrics, the appropriate technique to be used for the image is suggested.

```
Original image size: (178, 178, 3)
Interpolated image size: (712, 712, 3)
          Technique        PSNR      SSIM         MOS
1          Bilinear  361.201999  1.000000  181.101000
2           Bicubic   43.110840  0.993029   22.051934
4            Spline   43.110840  0.993029   22.051934
3           Lanczos   42.260622  0.989512   21.625067
0  Nearest Neighbor   34.241381  0.965385   17.603383
The best technique for the given image is Bilinear
```

(a)

```
Original image size: (301, 412, 3)
Interpolated image size: (451, 618, 3)
          Technique       PSNR      SSIM        MOS
2           Bicubic  41.237139  0.990827  21.113983
4            Spline  41.237139  0.990827  21.113983
3           Lanczos  41.111459  0.989617  21.050538
1          Bilinear  40.058107  0.990770  20.524439
0  Nearest Neighbor  30.933305  0.920616  15.926960
The best technique for the given image is Bicubic
```

(b)

Fig. 16. Comparison of the results

Figure 16(a) clearly depicts that , for the given Figure 8 , the Bilinear Interpolation technique best suites as compared to the other techniques.For Figure 12, Bicubic interpolation is the best choice

5.1. Advantages

The key advantage is identification of the best interpolation technique for a given image, which can lead to improved image quality and resolution. The use of various evaluation metrics to objectively compare the working (performance) of different interpolation techniques is also an added advantage.

5.2. Applications

A comparative study of different interpolation techniques for super-resolution is applicable in disparate fields like

- Radiology: to enhance the detail in imaging technologies for better diagnose and treat disease.
- Surveillance: to improve the quality of surveillance footage for better identification of objects and people.
- Satellite imaging: to increase the resolution of satellite images for better analysis and understanding of earth's surface.
- Photography and videography: to enhance the quality of images and videos captured by cameras, especially in low light conditions.

6. Conclusion

The paper effectively compared five different interpolation techniques: nearest neighbor, bilinear, bicubic, spline, and Lanczos. The parameters like advantages, disadvantages, speed, and suitability for different applications are clearly discussed. Nearest neighbor interpolation is the simplest and fastest interpolation technique, but it can produce jagged results. Bilinear interpolation is a more accurate interpolation technique than Nearest neighbor, but it is also slower and do not perform well with sharp edges. Bicubic interpolation is the most accurate interpolation technique, but it may produce blurred effects. Spline interpolation is a good compromise between accuracy and speed. Lanczos interpolation is a more accurate interpolation technique than Bicubic interpolation. The suitability of each interpolation technique depends on the specific application. In conclusion, there is no single "best" interpolation technique. The best technique for a particular application depends on the specific requirements of that application. To further enhance the visual quality, future studies might examine various interpolation methods, such as those based on deep learning

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